A077 – Electron Layer & Leapfrogging Unknots

Falsification Protocol and Report – v0.1.1

SST-Workbench – mathematical research track

28 October 2026

SCIENTIFIC STATUS: DEVELOPMENT / DIAGNOSTIC ONLY. Numerical ring-parity and a three-dimensional finite-width periodic Euler pilot do not prove an electron, quantized orbital layers, material vortex-tube stability, or spin 1/2. Original A077-v0.1.0 remains unmodified.

1 Identification, scope, and status

Catalog family `A077_Electron_Layer_Leapfrogging`; version `A077-v0.1.1`. This release is a new, standalone instance produced by the official Framework v1.1.0 generator, with an embedded hash-verified source snapshot. Its parent is A077-v0.1.0, whose runtime source had two missing files in a prior distribution; the new instance is generated from clean, complete release sources instead of copying a compromised snapshot. This project implements a reduced circular-filament control and an optional, separate periodic Fourier–Galerkin Euler pilot. Numerical evidence is not physical validation.

2 Falsifiable scientific question and objectives

Do unforced, mutually induced coaxial closed ring filaments exhibit resolution-consistent overtaking events without requiring discrete energy levels? Does adding an independent azimuthal (toroidal) velocity component to a finite-width, meridional (poloidal) periodic Euler flow generate an additional, demonstrable dynamical invariant or a physically derived spinorial response? *The second question remains unresolved by this first finite-width pilot.* Vertical ring-order permutations are never labeled electron levels $N \to N \pm 1$.

3 Competing hypotheses and decision rules

H_0: continuous classical ring coordinates, circulation-sign reversal, and smooth Euler velocity fields generate no intrinsically quantized spectrum or 2π sign-flipping spinor by themselves. H_1: a continuum-qualified material carrier yields independently stable, discrete intrinsic states and a non-postulated spinor-state representation whose 2π holonomy is -1 and whose 4π holonomy is $+1$. In v0.1.1, classical rank swaps are measured but do not test H_1. Absent independent physical representation, `G8 = UNRESOLVED`; a trivial 2π periodicity test of an ordinary vector field is not a spinor test.

4 Assumptions, applicability and excluded regimes

The target fluid is incompressible, inviscid and unforced with universal evolution time t. *Reduced-ring model:* ring centrelines are constrained to circles, cross-section follows an imposed $a_i^2 R_i$ proxy, the mutual induction kernel is softened, and the self-speed uses a thin-core closure. No stretching, field pressure or nonaxisymmetric modes are resolved. *Finite-width field pilot:* periodic cube, smooth core-scale data, Fourier–Galerkin 2/3 dealiasing, Leray projection and fourth-order Runge–Kutta (RK4). Vorticity is resolved on the chosen spectral grid but no topological/material core is identified and no continuum convergence is certified. Thin-core near-contact, unbounded-domain extrapolation, spinorial topology, photon emission and atmospheric plasma dynamics are outside the validated domain.

5 Notation, units and dimensional consistency

Symbol	Quantity	SI unit	Dimension
R_i, z_i, a_i	ring radius, axial position, core proxy	m	L
Γ_i	signed circulation	$\mathrm{m}^2\ \mathrm{s}^{-1}$	L^2T^{-1}
ρ_f	effective fluid density	$\mathrm{kg\ m}^{-3}$	ML^{-3}
t, t_0	absolute evolution time, chosen time scale	s	T
$\mathbf{u}, \boldsymbol{\omega}$	velocity, vorticity	$\mathrm{m\ s}^{-1}$, s^{-1}	LT^{-1}, T^{-1}
E, H	kinetic energy, kinematic helicity	J, $\mathrm{m}^4\ \mathrm{s}^{-2}$	ML^2T^{-2}, L^4T^{-2}
δ_s	sampled surface clearance	m	L

Conditional SST scaling, *not a fitted electron prediction*: $R_0 = r_c = 1.40897017 \times 10^{-15}\,\mathrm{m}$, $\|\mathbf{v}_\circlearrowleft\| = 1.09384563 \times 10^6\,\mathrm{m\ s}^{-1}$, $\Gamma_0 = 2\pi R_0\|\mathbf{v}_\circlearrowleft\|$ and $t_0 = R_0^2/\Gamma_0$. The ring code runs with R/R_0, z/R_0 and t/t_0; the Euler pilot is separately dimensionless with length R_0 and an explicitly chosen reference velocity, and no universal clock conversion is assumed.

6 Exact equations and derived measurements

E1 – regularized mutual induction. For circular source $\mathbf{X}_j(\varphi) = (R_j\cos\varphi, R_j\sin\varphi, z_j)$,

$$\mathbf{u}_{i\leftarrow j} = \frac{\Gamma_j}{4\pi}\int_0^{2\pi} \frac{\mathbf{X}_j'(\varphi)\times(\mathbf{x}_i - \mathbf{X}_j(\varphi))}{(\|\mathbf{x}_i - \mathbf{X}_j(\varphi)\|^2 + a_j^2)^{3/2}}\,d\varphi, \quad i \neq j. \tag{1}$$

This defined softened kernel is a proxy, not a resolved finite-core Euler Green function. $[\Gamma X' r/r^3] = L/T$.

E2 – slender-core closure and E3 – imposed core-volume proxy.

$$U_i^{\mathrm{self}} = \frac{\Gamma_i}{4\pi R_i}\left[\log\left(\frac{8R_i}{a_i}\right) - \frac{1}{4}\right], \tag{2}$$

$$a_i(t)^2 R_i(t) = a_i(0)^2 R_i(0), \qquad \dot{R}_i = \sum_{j\neq i} u_{R,i\leftarrow j}, \quad \dot{z}_i = U_i^{\mathrm{self}} + \sum_{j\neq i} u_{z,i\leftarrow j}. \tag{3}$$

The coefficient $1/4$ is a model convention. $[U_i] = L/T$; $[a_i^2 R_i] = L^3$.

E4–E6 – crossings, surface clearance and refined root times.

$$\Delta z(t) = z_2(t) - z_1(t), \qquad \Delta z(t_c) = 0 \text{ with a sign change}, \tag{4}$$

$$\delta_s(t) = \sqrt{(R_1 - R_2)^2 + (z_1 - z_2)^2} - (a_1 + a_2), \tag{5}$$

$$\Delta z_H(s) = (2s^3 - 3s^2 + 1)f_0 + (s^3 - 2s^2 + s)hf_0' \tag{6}$$

$$+ (-2s^3 + 3s^2)f_1 + (s^3 - s^2)hf_1', \quad s = (t - t_0)/h. \tag{7}$$

The crossing estimator solves $\Delta z_H(s) = 0$ only for sign-changing sampled brackets. This cubic interpolation is not exact RK4 dense output; root times require independent time-step convergence. An observed positive δ_s is a sampled circular-tube clearance proxy, not a guarantee of continuum core disjointness.

E7–E9 – finite-width periodic Euler pilot.

$$\partial_t \mathbf{u} = \mathcal{P}_{\mathcal{K}}(\mathbf{u}\times\boldsymbol{\omega}), \quad \boldsymbol{\omega} = \nabla\times\mathbf{u}, \quad \nabla\cdot\mathbf{u} = 0, \tag{8}$$

$$\mathbf{u}(0) = \mathcal{P}_{\mathcal{K}}\left[\nabla\times(A_\varphi\mathbf{e}_\varphi) + v_\varphi\mathbf{e}_\varphi\right], \tag{9}$$

$$E = \frac{\rho_f}{2}\int |\mathbf{u}|^2\,d^3x, \qquad H = \int \mathbf{u}\cdot\boldsymbol{\omega}\,d^3x. \tag{10}$$

The smooth Gaussian toroidal seed has independently specified meridional and azimuthal velocity amplitudes; v_φ is not a new circulation quantum. $\mathcal{P}_{\mathcal{K}}$ is the divergence-free Fourier projection with spectral cutoff. Dimensions are $[\partial_t\mathbf{u}] = [\mathbf{u}\times\boldsymbol{\omega}] = LT^{-2}$, $[E] = ML^2T^{-2}$, $[H] = L^4T^{-2}$. The spectral pilot uses normalized coordinates and diagnostics. Its discretized dynamics do not establish the continuum vorticity-knot topology.

Automatic equation register.

Machine-readable science contract

Research question Do regularized coaxial vortex-ring rank swaps and smooth independent meridional/azimuthal finite-width Euler initial fields yield convergent dynamics; is a new nonimposed spinorial two-turn response demonstrated?

Objective Separate classical geometric ring overtakings and numerical backend parity from true discrete energy levels, finite-core Euler carrier dynamics, and genuine spin-1/2 representation. v0.1.1 adds dense trace, Hermite root/event convergence, clearance, a periodic 3D Euler research pilot and independent numerical G5/G10 gates.

H_0 Continuous classical ring-overlap-free trajectories, sign-dependent velocity and smooth Galerkin Euler evolution do not in themselves produce quantized intrinsic energies, electron identity, or projective spinor holonomy.

H_1 An independently certified continuum finite-core Euler model exhibits discrete, parameter-robust intrinsic eigenclasses and uniquely derived spinorial 2pi/4pi holonomy without imposed state phase. This hypothesis is NOT evaluated in v0.1.1.

Registered equations

E1: regularized pairwise ring induction quadrature

$$u_{i<-j} = Gamma_j/(4pi)integral_0^{2pi}[X_j^prime(phi)cross(x_i - X_j(phi))]/(|x_i - X_j(phi)|^2 + a_j^2)^(3/2)dphi \quad (11)$$

Dimensional check: (L^2/T)*L*L/L^3=L/T

E2: slender-core self speed, explicit model choice

$$U_i = Gamma_i/(4piR_i)[ln(8R_i/a_i) - 1/4] \quad (12)$$

Dimensional check: (L^2/T)/L = L/T

E3: imposed thin-tube area-length proxy

$$a_i^2(t)R_i(t) = a_i^2(0)R_i(0) \quad (13)$$

Dimensional check: L^3=L^3

E4: axial-order crossing event

$$Deltaz(t) = z_2(t) - z_1(t); Deltaz(t_cross) = 0withsignreversal \quad (14)$$

Dimensional check: Delta z has L, t_cross has T

E5: coaxial minimal core tube clearance

$$d_surf = sqrt((R_1 - R_2)^2 + (z_1 - z_2)^2) - (a_1 + a_2) \quad (15)$$

Dimensional check: L-L=L

E6: cubic Hermite bracketed event estimator

$$Deltaz_h(s) = (2s^3 - 3s^2 + 1)f_0 + (s^3 - 2s^2 + s)hf_0prime + (-2s^3 + 3s^2)f_1 + (s^3 - s^2)hf_1prime \quad (16)$$

Dimensional check: f has L, h*fprime has L; s dimensionless

E7: periodic incompressible Euler Galerkin ODE

$$partial_t u = P_K(ucrossomega), omega = curlu, divu = 0 \tag{17}$$

Dimensional check: [u omega]=L/T^2=[du/dt]

E8: independent initial meridional and azimuthal velocity components

$$u_0 = P_K(curl(A_p hie_p hi) + v_p hie_p hi), withv_p hi = b * sum(r/R_i) * Gaussian_i \tag{18}$$

Dimensional check: [curl A]=L/T; [v_phi]=L/T, b dimensionless if velocity normalized

E9: Euler helicity and kinetic energy diagnostics

$$E = (rho_f/2)integral|u|^2 dV; H = intudotcurludV \tag{19}$$

Dimensional check: [E]=ML^2/T^2; [H]=L^4/T^2

Registered code-to-mathematics steps

1. **S1** [G1; equations E1]: Validate frozen framework and source input exact SHA-256; load analytic ring scenarios
2. **S2** [G2; equations E1, E2, E3]: Run full 401-sample reduced-ring RK4 trajectory and write raw CSV with conditional SI conversion
3. **S3** [G2; equations E4, E5, E6]: Locate bracketed cubic Hermite root crossings and minimum sampled tube clearance
4. **S4** [G2; equations E4, E6]: Independent ntheta and dt refinements at same fixed t_final; match event counts and root shifts
5. **S5** [G5; equations E1, E2]: Execute pybind11 science kernel and numeric reference in same instance; pass only if actual execution and parity
6. **S6** [G10; equations E1, E2]: After backend matrix framework assesses two genuinely executed backend categories and independent numeric parity
7. **S7** [G8; equations E7, E8, E9]: Run optional 3D Fourier Galerkin Euler research pilot, separate no-toroidal and + and - azimuthal swirl amplitudes
8. **S8** [G8; equations E7]: Do not score spinor without physically identified state representation, holonomy and held-out continuum convergence

7 Equation-to-code sequence and source mapping

1. Load and hash `configs/pilot.json` and its four disclosed analytic sign controls.
2. Evaluate E1–E3 using `science/ring_model.py`, write complete time series with all 401 samples of each case and conditional SI conversion.
3. Compute E4–E6 in `science/diagnostics.py`: bracketed cubic Hermite event roots, minimum centreline/core clearance, and three-level angular and temporal refinements at identical final time.
4. Independently execute native C++ via `native/cpp/native.cpp` and compare with Python, recording compiled module hash, source hash and common input hash. The Framework rechecks execution and numeric parity under G10.
5. Optionally, separately run `python -m science.euler_field` using `configs/euler_pilot.json`, generating three finite-width velocity-field cases: azimuthal amplitude 0, $+b$, and $-b$.
6. Record Euler energy, helicity, enstrophy, divergence, step size and spectral-grid configuration. Require future independent full spatial/temporal/core-width studies before any physical stability claim.
7. Preserve G3, G4 and G8 as unresolved. Any future reveal is a distinct private, preregistered operation, never part of diagnostics.

8 Provenance, sources and independence

The four ring controls are deterministic analytic input conditions, not four independent laboratory datasets. The ring quadrature and timestep levels are numerical refinements, not independent source families. The Euler Galerkin module is an independent *model class* from the reduced filament, not an independently validated material vortex-tube solver. No E010 knot atlas is copied into this instance, no physical electron mass/charge targets are used to fit the kernels, and no private reveal data enter public blind archives. The self-contained Framework v1.1.0 runtime source is pinned with its exact 142-file SHA-256 inventory; the frozen protocol has its own byte-exact manifest and commitments.

9 Gate dependency graph and scientific ceilings

Gate	Test and qualifying evidence	Max authority now
G0	Official frozen protocol and embedded source snapshot	TECHNICAL
G1	Analytic source/control hashes	TECHNICAL
G2	Angle/time refinement, Hermite events, checks; no continuum Euler	PROXY-ONLY
G3	Blind, held-out intrinsic spectral quantization	UNRESOLVED
G4	Independent full-field finite-core continuum confirmation	UNRESOLVED
G5	Executed native pybind11 ring kernel parity, independent of G4	NUMERIC-ONLY
G6	GPU science kernel, only if actually executed	NOT APPLICABLE
G7	External source-family replication	NOT APPLICABLE
G8	Euler continuum, intrinsic layers and spinorial bundle	UNRESOLVED
G9	Commitment-verified private reveal after qualified gates	DEFERRED
G10	Framework two-of-three executed backends plus numerical parity	NUMERIC-ONLY

G5 and G10 require only G2: actual numerical parity must not be causally blocked by unrelated physical G4. G8 continues to depend on genuinely independent physical G3/G4/G7. A G10 PASS is not a physical G8 PASS; the framework separates observed status from evidence qualification. The additional physical subtests P01–P07 are described by raw diagnostics and research limitations, not promoted to substitute framework gates.

10 Numerical refinements and predeclared thresholds

For $t_f/t_0 = 5$ and reference same-sign pair $R = (1, 0.8)$, $z = (-.15, +.15)$, $\Gamma = (+1, +1)$ and initial $a/R = .1$:

- Angular quadrature levels $n_\varphi = (48, 96, 192)$ at 400 RK4 steps.
- Temporal levels $(200, 400, 800)$ steps at $n_\varphi = 96$.
- Relative end-state tolerances 2×10^{-2} (angular), 10^{-2} (temporal); both crossing counts must agree and maximum paired root shift $\leq 2 \times 10^{-3}$ in t/t_0.
- Sampled clearance $\delta_s > 0$ is a separate domain check; its failure blocks a non-contact physical interpretation, even if the proxy numerical gate passes.
- Euler pilot uses $(n, L, t_f, \Delta t) = (24, 8, .02, .002)$ and only illustrates short-time bounded Galerkin evolution; no spatial, temporal or core convergence is certified. A future physical gate requires n-refinement and Δt-refinement at fixed physical end time, *plus* width, box and nonaxisymmetric perturbation studies.

Small endpoint differences and exact machine-level conservation can result from the Fourier–Galerkin truncation; neither establishes a physical restoring force, self-confinement or stable particle state.

11 Backend authority, cross-checks and Windows build

The Python NumPy binary64 code is the numerical reference. The independent native C++17 pybind11 ring kernel is accepted only after a genuine compile/execution, signed compiled binary/source hashes and ℓ_2 parity $\leq 10^{-10}$. Framework G10 further requires at least two real scientific categories from Python/C++/GPU. Intel Arc XPU and NVIDIA CUDA are a single GPU category, not two. Hardware preflight is not scientific execution. A missing C++/GPU implementation keeps FULL/CERTIFY unqualified; no silent Python fallback. DD32/FP32x2 remains `SCREENING_ONLY`. The optional periodic Euler pilot is currently Python-only, so native ring parity does not certify the independent field solver.

12 Blinding, private keys, and reveal isolation

The protocol contains no target-tuned electron mass, electric charge, Planck constant, measured spin projection, or discovered energy-level labels. The Framework creates new private opaque-ID and reveal-nonce bytes for this *new* version; they are stored under `private/`, never in a BLIND ZIP. The blind manifest, ledger, and source hashes are immutable once frozen. Reveal is blocked unless the independent scientific gate and commitment policy qualify it. Repeated BASIC/FULL/CERTIFY runs have unique identities and retain older outputs. The distributed source archive is private because it contains generated private credentials.

13 Null model, multiplicity and interpretation controls

The deterministic reduced-model trajectories have *no stochastic null ensemble*. Counting RK4 timesteps, sign configurations or grid levels as independent replicates is prohibited. The trial is not a statistical discovery of quantum levels; no p-values or multiple-testing claims are made. Future discrete-mode studies require blinded held-out parameters, a preregistered operator and continuum/finite-core null, familywise correction if multiple candidate modes are searched, and independent source confirmation.

14 Measured results and automatic immutable run provenance

Automatic run provenance

Field	Recorded value
Run ID	20261029T142919245662Z-676684b9
Run phase	BLIND
2-of-3 backend execution	EXECUTION_CONTRACT_PASS
Executed backend categories	py, cpp
GPU provider	intel
Protocol bundle SHA-256 (byte exact)	735cf572d6567e4b23828dda6e5876c58b4b74f598f1ccbf089816ea2d5fff03
Portable protocol diagnostic SHA-256	18142fa6c3236919fdd9e09e2ab1e282494d722f83ed04031108e2ef8c1edd80
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.13.9 (tags/v3.13.9:8183fa5, Oct 14 2025, 14:09:13) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Science evidence SHA-256	f2f766b11c66c7683103a2ddbc8bf303461b924bcc409bf79caef7abfa586fe8
Final output manifest	See OUTPUT_MANIFEST.json (finalized after PDF and report metadata)

Gate ledger

Gate	Status	Question	Reason
G0	PASS	Official frozen v1.1.0 runtime integrity?	Framework frozen protocol and embedded runtime verified before science.
G1	PASS	Analytic source/initial data provenance?	Frozen analytic case definitions and source hashes; not independent empirical observations.
G2	PASS	Reduced-model numerical and dimensional controls?	Separate angle-quadrature and ODE time-step convergence with event-root tests; no 3D Euler authority.
G3	UNRESOLVED	Target-free spectral discreteness rather than rank swaps?	No held-out eigenvalue spectrum or discrete level detection, geometric rank swaps are classical.
G4	UNRESOLVED	Independent full Euler field/held-out source confirmation?	No independent finite-core continuum Euler confirmation with space/time/core convergence.

G5	PASS	Does the actually executed independent C++ pybind kernel numerically agree with Python within the preregistered tolerance?	Actually executed instance pybind11 C++ ring science kernel; numeric parity only.
G6	NOT_APPLICABLE	Optional genuine GPU numerical screening (no GPU means NOT_APPLICABLE)?	Disabled by framework profile
G7	NOT_APPLICABLE	Independent source-family replication, not refinement/repeated run?	Disabled by framework profile
G8	UNRESOLVED	Do continuum full-Euler and spinorial mechanism gates close independently?	Independent toroidal/poloidal finite-width Euler pilot exists but is not continuum-certified; no spinor/field topology proof or experimental electron evidence.
G10	UNRESOLVED	Executed two of three genuinely independent science implementations with preregistered numerical parity?	No scientific evaluator implemented for this gate; not executed
G9	DEFERRED	Only after physical gates qualify, is private reveal allowed?	Reveal is a separate, commitment-verified phase and is not executed during blind runs.

Scientific qualification (separate from measured gate outcomes)

Gate	Evidence qualification	Blocking prerequisites
G0	QUALIFIED	none
G1	QUALIFIED	none
G2	QUALIFIED	none
G3	UNRESOLVED	none
G4	DIAGNOSTIC_ONLY	G3
G5	QUALIFIED	none
G6	NOT_APPLICABLE	none
G7	NOT_APPLICABLE	G4
G8	DIAGNOSTIC_ONLY	G3, G4, G7
G10	UNRESOLVED	none
G9	DEFERRED	G8

Numeric backend comparisons (not physical validation)

Backend	Relative L2 error	Result/status
cpp	5.923075954129776e-16	MEASURED_ONLY

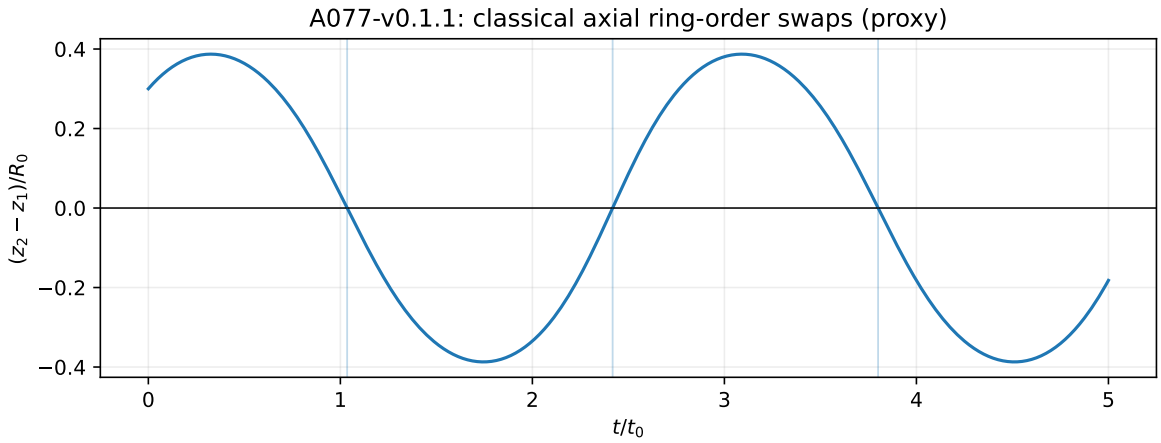
Backend manifest


```
{
  "backend_2of3": {
    "artifact": "AUTO_BACKEND_MATRIX.json",
    "counted": 2,
    "executed": [
      "py",
      "cpp"
    ],
    "status": "EXECUTION_CONTRACT_PASS"
  },
  "model": "AXISYMMETRIC_REGULARIZED_FILAMENT_PROXY",
  "note": "G5 local pybind check independent of G4. G10 validates
multi-backend parity after execution. No physical spin/level pass.",
  "schema": "SST-BACKEND-MANIFEST-2",
  "science_diagnostics": "science_diagnostics/scientific_diagnostics.json",
  "scientific_authority": "DEVELOPMENT_DIAGNOSTIC_ONLY"
}
```


Source manifest


```
{
  "manifest_sha256":
```

```
"bee7a88039d6c7375b5ec2d8eb062c1766a9113ff51c2a15c5290d27e4d8841d",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [],
  "workbench_root": "C:\\workspace\\projects\\SST-Workbench"
}
```

15 Scientific interpretation, falsifiers and minimal experiments

ORTHODOX: Euler circulation transport and Biot–Savart velocity reconstruction are classical conditional results. **DERIVED UNDER ASSUMPTIONS:** ring trajectory roots, clearance proxies and spectral Euler short-time diagnostics are numerical statements within their explicitly discretized models. **SPECULATIVE–RESEARCH–TRACK:** identification of CW/CCW, ring order, independent toroidal/meridional velocity, jet/sprite morphology or leapfrogging phases with electron spin-1/2 or physical quantum layers. A real velocity vector field is single-valued under a 2π spatial rotation; that does *not* supply $\Psi(2\pi) = -\Psi(0)$ for a derived physical state, nor the exchange sign of two identical fermions. Future minimal tests: (i) fixed-time spatial/time/core refinement for the field pilot, (ii) helicity and link invariants on genuine disjoint material tubes, (iii) independent survival/repetition of internal modes, (iv) configuration-space rotation and braid loop analysis without assuming a spinorial bundle. No sprite/giant-jet ionization or electromagnetic emission is modeled.

16 Exact execution, outputs and stop criteria

```
cd /d <EXTRACTED_A077-v0.1.1>
python -m pytest -q tests
run_all.cmd BACKEND_PREFLIGHT
run_all.cmd FREEZE
run_all.cmd BASIC
run_all.cmd FULL
run_all.cmd CERTIFY
python -m science.euler_field ^
  --config configs/euler_pilot.json ^
  --output A077_euler_pilot_v0.1.1-outputs
```

In `A077-v0.1.1` the Framework writes `electron_layer_leapfrogging_v0.1.1-outputs/`, preserving earlier runs and publishing an immutable run-specific BLIND ZIP and its SHA-256 sidecar in the parent. Source archive must be extracted into *new* version directory only. Halt physical certification on failed source hashes, absent native science backend, Euler-grid/core nonconvergence, missing evidence, frozen protocol mutation, core contact, or missing spinorial state definition. No user hardware result is claimed in advance.

17 References

References

[1] H. Helmholtz (1858), "On Integrals of the Hydrodynamical Equations Which Express Vortex Motion," *Journal für die reine und angewandte Mathematik* 55, 25–55. DOI: `https://doi.org/10.1515/crll.1858.55.25`.

[2] P. G. Saffman (1992), *Vortex Dynamics*, Cambridge University Press. DOI: https://doi.org/10.1017/CBO9780511624063.

[3] J. Dávila, M. del Pino, M. Musso, J. Wei (2024), "Leapfrogging vortex rings for the three-dimensional incompressible Euler equations," *Communications on Pure and Applied Mathematics* 77, 3843–3957. DOI: https://doi.org/10.1002/cpa.22199.

[4] A. J. Niemi (2005), "Exotic Statistics of Leapfrogging Vortex Rings," *Physical Review Letters* 94, 124502. DOI: https://doi.org/10.1103/PhysRevLett.94.124502.

[5] D. Finkelstein and J. Rubinstein (1968), "Connection between Spin, Statistics, and Kinks," *Journal of Mathematical Physics* 9, 1762–1779. DOI: https://doi.org/10.1063/1.1664510.

[6] S. A. Orszag (1971), "On the elimination of aliasing in finite-difference schemes by filtering high-wavenumber components," *Journal of the Atmospheric Sciences* 28, 1074. DOI: https://doi.org/10.1175/1520-0469(1971)028<1074:OTEOAI>2.0.CO;2.

[7] T. J. Dekker (1971), "A Floating-Point Technique for Extending the Available Precision," *Numerische Mathematik* 18, 224–242. DOI: https://doi.org/10.1007/BF01397083.